SAMPLE COMPUTATIONS IN ASS: THE DIFFERENTIALS ON h_j^3

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Abstract. These notes are for the talks in the steenrod algebra seminar at Peking university during 2026 summer. In these talks we will make use of the motivic cofiber of τ and $\mathbb{F}_2$-synthetic homotopy theory and explain the sample computations of [BX25]. We will also explain the use of synthetic power operations in this sample computation.

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Introduction

The Adams spectral sequence and Adams-Novikov spectral sequence are powerful computational tools in stable homotopy theory, which help us to convert the extremely difficult task to compute the stable homotopy group into the very difficult task to compute certain pages and differentials. In these talks, we will explain the computation of [BX25]. The main theorem is:

Theorem 0.1 (Theorem 1.11; [BX25])**.** *$d_4(h_j^3) = h_0^3 g_{j-2} \neq 0$ for $j \geq 6$.*

Our main strategy is to use the Miller square to analyze the E_2 page and use motivic cofiber of τ zigzag to analyze higher pages and differentials.

Throughout these talks, we live in p-complete world and we require $p = 2$ most of time.

1. Recollection of Miller square and mot-ASS

1.1. **The Miller square.** Consider the (classical) Adams spectral sequence and the Adams-Novikov spectral sequence:

$$E_2^{s,t} \cong \mathrm{Ext}_{\mathcal{A}}^{s,t}(\mathbb{F}_p, \mathbb{F}_p) \Longrightarrow \pi_{t-s}S^0 \quad d_r \colon E_r^{s,t} \to E_r^{s+r,t-r+1}$$
$$E_2^{s,t} \cong \mathrm{Ext}_{\mathrm{BP}_*\mathrm{BP}}^{s,t}(\mathrm{BP}_*, \mathrm{BP}_*) \Longrightarrow \pi_{t-s}S^0 \quad d_r \colon E_r^{s,t} \to E_r^{s+r,t-r+1}$$

The Miller square brings a way to compare these two spectral sequences on the E_2 page. It consists of the above two spectral sequences, the other two spectral sequences computing them, and an identification of the E_2 pages.

Recall that $\mathrm{BP}_* \cong \mathbb{Z}_{(p)}[v_0, \cdots]$. We can use the augmented ideal $I = (v_0, \cdots, v_n, \cdots)$ to give a purely algebraic filtration on BP^*. This leads to a tri-indices spectral sequence

$$E_2^{s,k,t'} \cong \mathrm{Ext}^{s,t'}_{\mathrm{BP}_*\mathrm{BP}/I}(\mathrm{BP}_*/I, I^k/I^{k+1}) \Longrightarrow \mathrm{Ext}^{s,t'}_{\mathrm{BP}_*\mathrm{BP}}(\mathrm{BP}_*, \mathrm{BP}_*)$$

$$d_r \colon E_r^{s,k,t'} \to E_r^{s+1.k+r-1,t'}$$

which we refer to as the *algebraic Novikov spectral sequence*.

On the other hand, fix $p = 2$, we have the subalgebra of the steenrod algebra:

$$\mathcal{P} = \mathbb{F}_2[\xi_1^2, \cdots, \xi_n^2, \cdots] \subset F_2[\xi_1, \cdots, \xi_n, \cdots] = \mathcal{A}$$

and the induced quotient gives the exterior part $\mathcal{Q} \cong \Lambda[\xi_1, \cdots, \xi_n, \cdots]$. Therefore, associated with the extension of Hopf algebra $\mathcal{P} \to \mathcal{A} \to \mathcal{Q}$, we have a Cartan-Eilenberg spectral sequence:

$$E_2^{s,k,t} = \mathrm{Ext}^{s,t}_{\mathcal{P}}(\mathbb{F}_2, \mathrm{Ext}^{k}_{\mathcal{Q}}(\mathbb{F}_2, \mathbb{F}_2)) \Longrightarrow \mathrm{Ext}^{s+k,t}_{\mathcal{A}}(\mathbb{F}_2, \mathbb{F}_2)$$

$$d_r \colon E_r^{s,k,t} \to E_r^{s+t.k+r-1,t}$$

The above two second pages are identified. To see this, we first note there is a Hopf algebra identification.

Lemma 1.1. *There is a Hopf algebra identification* $(\mathbb{F}_2, \mathcal{A}) \cong (\mathrm{BP}_*/I, \mathrm{BP}_*\mathrm{BP}/I)$.

Proof. The Hopf algebroid structure for $(\mathrm{BP}_*, \mathrm{BP}_*\mathrm{BP})$ can be found in [Rav03, Theorem 4.1.19], while after mod I it becomes a Hopf algebra and is equivalent to $(\mathbb{F}_2, \mathcal{A})$. □

Lemma 1.2. *We have an equivalence of graded algebras:*

$$\mathrm{Ext}^*_{\mathcal{Q}}(\mathbb{F}_2, \mathbb{F}_2) \cong \mathbb{F}_2[q_0, q_1, \cdots]$$

where the (k, t) *degree of* q_i *is* $(1, 2^{i+1} - 1)$

Proof. Since Q is primitively generated, the cobar complex that calculates $\mathrm{Ext}^*_{\mathcal{Q}}(\mathbb{F}_2, \mathbb{F}_2)$ is easy to compute. It is generated by $q_i = [\xi_{i+1}] \in \mathrm{Cobar}^1_{\mathcal{A}}(\mathbb{F}_2)$. The degree follows immediately. □

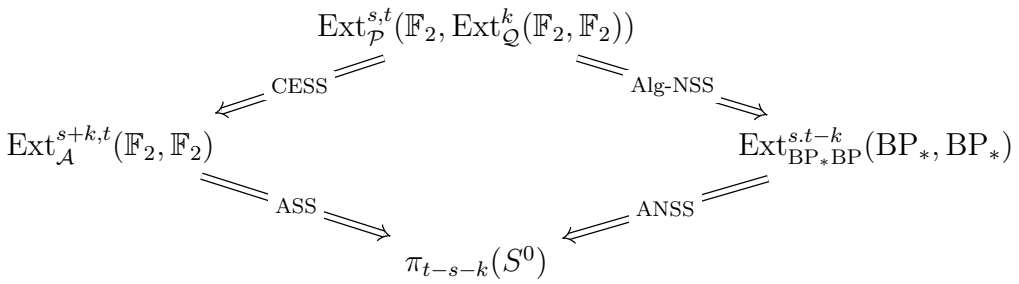
With these two lemmas in hand, we can identify the E_2-pages of Cartan-Eilenberg spectral sequence and algebraic Novikov spectral sequence. Basically, we have:

$$\mathrm{Ext}^*_{\mathcal{Q}}(\mathbb{F}_2, \mathbb{F}_2) \cong \mathbb{F}_2[q_0, q_1, \cdots] \cong \mathbb{F}_2[v_0, v_1, \cdots] \cong \bigoplus I^*/I^{*+1}$$

Here we naively identify q_i with v_i. Since q_i sits in (s, k, t)-degree $(0, 1, 2^{i+1} - 1)$ while v_i sits in (s, k, t') degree $(0, 1, 2^{i+1} - 2)$, we need to twist the degree slightly. As a result, we have an identification:

$$\mathrm{Ext}^{s,t}_{\mathcal{P}}(\mathbb{F}_2, \mathrm{Ext}^{k}_{\mathcal{Q}}(\mathbb{F}_2, \mathbb{F}_2)) \xrightarrow{\cong} \mathrm{Ext}^{s,t'}_{\mathrm{BP}_*\mathrm{BP}/I}(\mathrm{BP}_*/I, I^k/I^{k+1})$$

by sending s to s, k to k and t to $t' + k$. Altogether, we have the following Miller square:



The Miller square is convenient for us to transport d_2 differentials form alg-NSS to ASS through CESS. However, there are some limitations:

- The Miller square is totally algebraic and lives in spectral sequence level. That means, it inevitably loses the filtration data.
- As a result, we cannot read out information about higher differentials and higher pages from the Miller square.

In order to transport the data from higher pages of alg-NSS to ASS, we need to consider some deformation of classical Adams spectral sequence. That's where the motivic homotopy theory come in.

1.2. **The motivic Adams spectral sequence.** We briefly recall the motivic homotopy theory and the motivic Adams spectral sequence. We fix $k = \mathbb{C}$ and work within $\mathrm{SH}(\mathbb{C})$. We will denote $S^{k,w} = S^{k-w} \wedge \mathbb{G}_m^w$ to be the bigraded sphere. This convention is very convenient when we use Betti realization to compare $\mathbb{C}$-motivic spectra with classical ones.

More precisely. Suppose that $X \in \mathrm{SH}(\mathbb{C}) = \mathrm{Shv}_{\mathrm{Nis}}^{\mathbb{A}^1}(\mathrm{Sm}_{\mathbb{C}}, \mathrm{Sp})$, levelwise taking the $\mathbb{C}$-points gives the *Betti realization* functor:

$$\mathrm{Re}\colon \mathrm{SH}(\mathbb{C}) \to \mathrm{Sp}$$
$$X \mapsto X(\mathbb{C})$$

This functor sends the motivic sphere spectra $S^{k,w}$ to S^k, the motivic Eilenberg-Macline spectra $\mathrm{M}\mathbb{Z}/l$ to $\mathrm{H}\mathbb{Z}/l$, and the motivic Brown-Peterson spectra BPGL (see [Vez01] for the original construction) to BP.

Voevodsky's resolution of the Bloch-Kato conjecture gives an isomorphism (see [MVW06, Lecture 5] for details):

$$\mathrm{H}_{\mathrm{mot}}^{*,*}(\mathrm{Spec}(F), A) \cong \mathrm{K}_*^{\mathrm{M}}(F) \otimes_{\mathbb{Z}} A$$

Moreover, suppose that we use $\mathbb{M}_p$ to denote the $\mathbb{C}$-motivic cohomology of a point with $\mathbb{F}_p$-coefficience, we have isomorphisms:

$$\mathbb{M}_p \cong \mathrm{K}_*^{\mathrm{M}}(\mathbb{C})/p \cong \mathbb{F}_p[\tau]$$

with τ in bidegree $(0, -1)$.

The generator τ in motivic cohomology detects an element in $\pi_{0,1}(S)$ via mot-ASS. In fact, it is even in the image of the motivic Hurewicz map, as described in [HKO11]. This gives rise to an element τ in the motivic stem $\pi_{0,-1}$. This element is sent to unit by Betti realization.

We consider the fiber sequence

$$S^{0,-1} \longrightarrow S^{0,0} \longrightarrow S^{0,0}/\tau \longrightarrow S^{1,-1}$$

The work of Gheorghe [Ghe18] shows that $S^{0,0}/\tau$ becomes an $\mathbb{E}_\infty$-algebra in $\mathrm{SH}(\mathbb{C})$ with the unit map $S^{0,0} \to S^{0,0}/\tau$.

The computation of [Voe03] and [HKØ17] gives the following result of dual motivic steenrod algebra for $k = \mathbb{C}$.

$$\mathcal{A}^{\mathrm{mot}} \cong \mathbb{F}_2[\tau][\tau_1, \cdots][\xi_i, \cdots]/\tau_i^2 = \tau\xi_i$$

Using this, we formally construct the motivic Adams spectral sequence.

$$E_2^{a,t,w} = \mathrm{Ext}^{a,t,w}_{\mathcal{A}^{\mathrm{mot}}}(\mathbb{F}_p[\tau], \mathbb{F}_p[\tau]) \Longrightarrow \pi_{t-a,w}S^{0,0}$$
$$d_r \colon E_r^{a,t,w} \to E_r^{a+r,t+r-1,w}$$

The map $S^{0,0} \to S^{0,0}/\tau$ gives rise to a comparison of spectral sequences:

$$\mathrm{Ext}^{a,t,w}_{\mathcal{A}^{\mathrm{mot}}}(\mathbb{F}_p[\tau], \mathbb{F}_p[\tau]) \to \mathrm{Ext}^{a,t,w}_{\mathrm{mot}}(\mathbb{F}_p[\tau], \mathbb{F}_p)$$

The latter is closely related to the algebraic Novikov spectral sequence.

Theorem 1.3 ([GWX21]; Theorem 1.17)**.** *There is an isomorphism of tri-graded spectral sequences between the motivic Adams spectral sequence for $S^{0,0}/\tau$ and the algebraic Novikov spectral sequence.*

$$\begin{array}{ccc}
\mathrm{Ext}^{s+k,t'-k,\frac{t'}{2}}_{\mathcal{A}^{\mathrm{mot}}}(\mathbb{F}_p[\tau], \mathbb{F}_p) & \xrightarrow{\cong} & \mathrm{Ext}^{s,t'}_{\mathrm{BP}_*\mathrm{BP}/I}(\mathbb{F}_p, I^k/I^{k+1}) \\
\Vert & & \Vert \\
\Downarrow{\scriptstyle \text{mot-ASS}} & & \Downarrow{\scriptstyle \text{Alg-NSS}} \\
\pi_{t'-s,\frac{t'}{2}}(S^{0,0}/\tau) & \xrightarrow{\cong} & \mathrm{Ext}^{s,t'}_{\mathrm{BP}^*\mathrm{BP}}(\mathrm{BP}_*, \mathrm{BP}_*)
\end{array}$$

Remark 1.4. This identification makes sense: the E_2 page of the algebraic Novikov spectral sequence concentrates in even t' degree. By definition, this is equivalent to say the E_2 page of CESS $\mathrm{Ext}^{s,t}_{\mathcal{P}}(\mathbb{F}_2, \mathrm{Ext}^k_{\mathcal{Q}}(\mathbb{F}_2, \mathbb{F}_2))$ vanish when $t + k$ is odd. This is because we have the identification:

$$\mathrm{Ext}^{s,t}_{\mathcal{P}}(\mathbb{F}_2, \mathbb{F}_2) \cong \mathrm{Ext}^{s,2t}_{\mathcal{A}}(\mathbb{F}_2, \mathbb{F}_2)$$

Therefore, all generators of $\mathrm{Ext}^{s,t}_{\mathcal{P}}(\mathbb{F}_2, \mathrm{Ext}^k_{\mathcal{Q}}(\mathbb{F}_2, \mathbb{F}_2))$ satisfies that $t + k$ is odd.

As a consequence, we have the following motivic zigzag square:

$$\begin{array}{ccccc}
\mathrm{Ext}^{a,t}_{\mathcal{A}}(\mathbb{F}_p, \mathbb{F}_p) & \xleftarrow{\tau=1} & \mathrm{Ext}^{a,t,w}_{\mathcal{A}^{\mathrm{mot}}}(\mathbb{F}_p[\tau], \mathbb{F}_p[\tau]) & \xrightarrow{\tau=0} & \mathrm{Ext}^{a,t,w}_{\mathcal{A}^{\mathrm{mot}}}(\mathbb{F}_p[\tau], \mathbb{F}_p) \\
\Vert & & \Vert & & \Vert \\
\Downarrow{\scriptstyle \text{ASS}} & & \Downarrow{\scriptstyle \text{mot-ASS}} & & \Downarrow{\scriptstyle \text{mot-ASS}} \\
\pi_{t-a}(S) & \xleftarrow{\mathrm{Re}} & \pi_{t-a,w}(S^{0,0}) & \xrightarrow{\mathrm{can}} & \pi_{t-a,w}(S^{0,0}/\tau)
\end{array}$$

Philosophically, in order to read out additional data for classical Adams spectral sequence, we have the following procedure:

- We know partial data about the Adams spectral sequence and partial data about the algebraic Novikov spectral sequence.
- Using Theorem 1.3, we convert the partial data about the algebraic Novikov spectral sequence to the partial data about the motivic Adams spectral sequence for $S^{0,0}/\tau$.

- Using the motivic zigzag, we piece the partial data about the Adams spectral sequence and the partial data about the motivic Adams spectral sequence for $S^{0,0}/\tau$ together, obtaining "more than partial" data about the motivic Adams spectral sequence for $S^{0,0}$.
- Using Betti realization, we convert the "more than partial" data about the motivic Adams spectral sequence for $S^{0,0}$ to "more than partial" data about the classical Adams spectral sequence.

2. The main theorem

2.1. **Break the main theorem.** The motivic zigzag allows us to break down the task of understanding the classical Adams spectral sequence into three tasks: understanding the partial data of the classical Adams spectral sequence, the motivic Adams spectral sequence of sphere and the motivic Adams spectral sequence of cofiber of τ.

On the other hand, in order to obtain these partial data, we need to understand both the E_2-pages and the higher differentials. We treat these data with different strategies:

- For the E_2 pages, we will introduce a new spectral sequence to compute the E_2-page of CESS. The data about the E_2-page of ASS and mot-ASS for cofiber of τ can be read out from CESS. Then using the rigidity to obtain the data of the mot-ASS for sphere.
- For the differentials of the mot-ASS for cofiber of τ, we make use of Theorem 1.3, converting to differentials of alg-NSS. Then we descent to the cobar complex level to compute the desired differentials.
- For the differentials of the classical ASS, we will use $\mathbb{F}_2$-synthetic homotopy theory.

These three parts corresponds to the following three theorems:

Theorem 2.1. *Let $j \geq 6$.*

(1) The E_2 page of the classical ASS takes the following form near h_j^3:

$(a,\ \ t-a)$	$\mathrm{Ext}_{\mathcal{A}}^{a,t}(\mathbb{F}_2, \mathbb{F}_2)$
$(4,\ \ 3\cdot 2^j-4)$	$\mathbb{F}_2\{g_{j-2}\}$
$(5,\ \ 3\cdot 2^j-4)$	$\mathbb{F}_2\{h_0 g_{j-2}\}$
$(6,\ \ 3\cdot 2^j-4)$	*contains* $\mathbb{F}_2\{h_0^2 g_{j-2}\}$
$(7,\ \ 3\cdot 2^j-4)$	*contains* $\mathbb{F}_2\{h_0^3 g_{j-2}\}$
$(3,\ \ 3\cdot 2^j-3)$	$\mathbb{F}_2\{h_j^3\}$
$(4,\ \ 3\cdot 2^j-3)$	$\mathbb{F}_2\{h_0 h_j^3\}$
$(5,\ \ 3\cdot 2^j-3)$	$\mathbb{F}_2\{h_0^2 h_j^3, h_1 g_{j-2}\}$

with the following two exceptions:

- *When $j=6$, the degree $(a,t-a)=(5,3\cdot 2^6-4)$ contains an additional class $h_7 D_3(0)$.*
- *When $j=7$, the degree $(a,t-a)=(5,3\cdot 2^7-3)$ contains an additional class $h_8 D_3(1)$.*

(2) The E_2 page of the motivic ASS for sphere takes the following form near h_j^3:

$(a,\ \ t-a)$	$\mathrm{Ext}^{a,t,w}_{\mathcal{A}^{\mathrm{mot}}}(\mathbb{F}_2[\tau],\, \mathbb{F}_2[\tau])$
$(4,\ \ 3\cdot 2^j-4)$	$\mathbb{F}_2[\tau]\{g_{j-2}\}$
$(5,\ \ 3\cdot 2^j-4)$	$\mathbb{F}_2[\tau]\{h_0 g_{j-2}\}$
$(6,\ \ 3\cdot 2^j-4)$	*contains* $\mathbb{F}_2[\tau]\{h_0^2 g_{j-2}\}$, *$\tau$-torsion free*
$(7,\ \ 3\cdot 2^j-4)$	*contains* $\mathbb{F}_2[\tau]\{h_0^3 g_{j-2}\}$, *$\tau$-torsion free*
$(3,\ \ 3\cdot 2^j-3)$	$\mathbb{F}_2[\tau]\{h_j^3\}$
$(4,\ \ 3\cdot 2^j-3)$	$\mathbb{F}_2[\tau]\{h_0 h_j^3\}$
$(5,\ \ 3\cdot 2^j-3)$	$\mathbb{F}_2[\tau]\{h_0^2 h_j^3, h_1 g_{j-2}\}$

where each of the generators $g_{j-2}, h_0 g_{j-2}, h_0^2 g_{j-2}, h_0^3 g_{j-2}, h_j^3, h_0 h_j^3, h_0^2 h_j^3$ *has the same weight* $w = 3\cdot 2^{j-1}$ *and* $h_1 g_{j-2}$ *has the weight* $w = 3\cdot 2^{j-1}+1$. *There are following two exceptions:*

- *When* $j=6$, *the degree* $(a, t-a) = (5, 3\cdot 2^6-4)$ *contains an additional generator* $h_7 D_3(0)$.
- *When* $j=7$, *the degree* $(a, t-a) = (5, 3\cdot 2^7-3)$ *contains an additional generator* $h_8 D_3(1)$ *with weight* $w = 3\cdot 2^6+1$.

(3) The E_2 *page of the motivic ASS for cofiber of* τ *takes the following form near* h_j^3*:*

$(a,\ \ 2w-t+a,\ \ t-a)$	$\mathrm{Ext}^{a,t,w}_{\mathcal{A}^{\mathrm{mot}}}(\mathbb{F}_2[\tau],\, \mathbb{F}_2)$
$(4,\ \ 4,\ \ 3\cdot 2^j-4)$	$\mathbb{F}_2\{g_{j-2}\}$
$(5,\ \ 4,\ \ 3\cdot 2^j-4)$	$\mathbb{F}_2\{h_0 g_{j-2}\}$
$(6,\ \ 4,\ \ 3\cdot 2^j-4)$	*contains* $\mathbb{F}_2\{h_0^2 g_{j-2}\}$
$(7,\ \ 4,\ \ 3\cdot 2^j-4)$	*contains* $\mathbb{F}_2\{h_0^3 g_{j-2}\}$
$(3,\ \ 1,\ \ 3\cdot 2^j-3)$	0
$(4,\ \ 1,\ \ 3\cdot 2^j-3)$	0
$(5,\ \ 1,\ \ 3\cdot 2^j-3)$	0
$(6,\ \ 1,\ \ 3\cdot 2^j-3)$	0
$(3,\ \ 3,\ \ 3\cdot 2^j-3)$	$\mathbb{F}_2\{h_j^3\}$
$(4,\ \ 3,\ \ 3\cdot 2^j-3)$	$\mathbb{F}_2\{h_0 h_j^3\}$
$(5,\ \ 3,\ \ 3\cdot 2^j-3)$	$\mathbb{F}_2\{h_0^2 h_j^3\}$
$(6,\ \ 3,\ \ 3\cdot 2^j-3)$	$\mathbb{F}_2\{h_0^3 h_j^3\}$ *or* 0
$(5,\ \ 5,\ \ 3\cdot 2^j-3)$	$\mathbb{F}_2\{h_1 g_{j-2}\}$

The weights of generators are the same as that of (2). There are two exceptions:

- *When* $j=6$, *the degree* $(a, 2w-t+a, t-a) = (5, 4, 3\cdot 2^6-4)$ *possibly contains an additional class* $h_7 D_3(0)$.
- *When* $j=7$, *the degree* $(a, 2w-t+a, t-a) = (5, 5, 3\cdot 2^7-3)$ *contains an additional class* $h_8 D_3(1)$.

(4) The Betti realization and reduction mod τ *sends each generator to generators of the same name.*

Theorem 2.2. *Let* $j \geq 6$. *In the motivic Adams spectral sequence for cofiber of* τ, *we have the followings:*

(1) $d_2(h_j^3) = 0$.

(2) $d_3(h_j^3) = 0$.
(3) $d_4(h_j^3) = h_0^3 g_{j-2}$.

Theorem 2.3. *Let $j \geq 6$. In the classical Adams spectral sequence, we have the followings:*
(1) $d_2(h_j^3) = 0$.
(2) $d_3(h_j^3)$ *is either* 0 *or* $h_0^2 g_{j-2}$.
(3) $d_4(h_j^3)$ *is either* 0 *or* $h_0^3 g_{j-2}$ *if defined.*

Combining these three theorems together, we obtain the main theorem:

Theorem 2.4. *Let $j \geq 6$.*
(1) The class h_j^3 survives to E_4 page of classical ASS.
(2) $d_4(h_j^3) = h_0^3 g_{j-2} \neq 0$.

Proof. We use the data in Theorem 2.1, Theorem 2.2 and Theorem 2.3 step by step to deduce the consequence. We have:

- Using the first statement (1) of Theorem 2.3, we see that h_j^3 survives to E_3 page.
- By reading the tables of Theorem 2.1, we found that the only possible source for d_2 entering $h_0^2 g_{j-2}$ is $h_0 h_j^3$ in the classical ASS (also motivic ASS for spheres). The element is sent to zero by d_2. This implies that we have an injection induced by the Betti realization on the degree of $h_0^2 g_{j-2}$.
- Therefore, using the second claim of Theorem 2.3, in the mot-ASS, we also have that $d_3(h_j^3)$ is either 0 or $h_0^2 g_{j-2}$.
- However, we know that $d_3(h_j^3) = 0$ in the mot-ASS for the cofiber of τ from Theorem 2.2, meaning that $d_3(h_j^3)$ is either 0 or τ-multiples in mot-ASS for spheres. Therefore, we conclude that $d_3(h_j^3) = 0$ in mot-ASS for spheres.
- Using Betti realization, $d_3(h_j^3) = 0$ in classical ASS for spheres. That means, h_j^3 survives to E_4.
- For the d_4 differentials, a similar argument allows us to conclude that in the mot-ASS. we also have that $d_4(h_j^3)$ is either zero or $h_0^3 g_{j-2}$.
- However, we know that $d_4(h_j^3) = h_0^3 g_{j-2}$ in the mot-ASS for cofiber of τ. So unless $h_0^3 g_{j-2}$ itself is zero, otherwise we must have $d_4(h_j^3) = h_0^3 g_{j-2} \neq 0$ in mot-ASS.
- $h_0^3 g_{j-2}$ can not be zero in E_4. It survives at least to E_4. First, by reading the table for E_2 in Theorem 2.1, we see that the only possible source for d_2 differential entering $h_0^3 g_{j-2}$ is $h_0^2 h_j^3$, which is sent to zero by Leibnitz rule. For the same reason there is also no entering d_3-differentials.
- Therefore, one conclude that $d_4(h_j^3) = h_0^3 g_{j-2} \neq 0$ in mot-ASS. Using Betti realization, we have $d_4(h_j^3) = h_0^3 g_{j-2} \neq 0$ in classical ASS. □

Our next aim is to prove Theorem 2.1, Theorem 2.3 and Theorem 2.2.

2.2. **Determine the E_2 pages.** This subsection is devoted to prove Theorem 2.1. As explained in Theorem 1.3, the E_2 page of the motivic Adams spectral sequence for $S^{0,0}/\tau$ is isomorphic to the E_2 page of CESS. On the other hand, we can also read out the classical Adams E_2 page from CESS, and use rigidity to deduce the E_2 page of mot-ASS. All in all the upshot is to understand the CESS.

Using the identification $\mathrm{Ext}_{\mathcal{P}}^{s,2t}(\mathbb{F}_2, \mathbb{F}_2) \cong \mathrm{Ext}_{\mathcal{A}}^{s,t}(\mathbb{F}_2, \mathbb{F}_2)$, and the injective map

$$\mathrm{Ext}_{\mathcal{P}}^{s,t}(\mathbb{F}_2, \mathbb{F}_2) \to \mathrm{Ext}_{\mathcal{P}}^{s,t}(\mathbb{F}_2, \mathbb{F}_2[q_0, \cdots]) \cong \mathrm{Ext}_{\mathcal{P}}^{s,t}(\mathbb{F}_2, \mathrm{Ext}_{\mathcal{Q}}^{k}(\mathbb{F}_2, \mathbb{F}_2))$$

we can use similar name in classical ASS to denote elements in CESS. Note that the element $a \in \mathrm{Ext}_{\mathcal{P}}^{s,t}(\mathbb{F}_2, \mathbb{F}_2)$ detects $\mathrm{Sq}^0(a)$ in $\mathrm{Ext}_{\mathcal{A}}^{s,t}(\mathbb{F}_2, \mathbb{F}_2)$ through CESS by degree reasons.

In order to analyze the CESS, we introduce another spectral sequence that converge to its E_2 page, namely the *algebraic Atiyah-Hirzebruch spectral sequence*.

Construction 2.5. We give $\mathbb{F}_2[q_0, \cdots]$ a grading: we associate q_i with grade i. This gives a filtration on $\mathbb{F}_2[q_0, \cdots]$. Recall that the $\mathcal{P}$-comodule action on $\mathbb{F}_2[q_0, \cdots]$ is given by:

$$\psi(q_n) = \sum_{i=0}^{n} \xi_{n-i}^{2^{i+1}} \otimes q_i$$

Therefore the associated graded pieces are trivial $\mathcal{P}$-comodules. This results in a spectral sequence

$$E_1 \cong \mathrm{Ext}_{\mathcal{P}}(\mathbb{F}_2, \mathbb{F}_2) \otimes \mathbb{F}_2[q_0, \cdots] \Longrightarrow \mathrm{Ext}_{\mathcal{P}}(\mathbb{F}_2, \mathbb{F}_2[q_0, \cdots])$$
$$d_r \colon E_r^{s,t,k,i} \to E_r^{s+1,k,t,i-r}$$

where the class in $\mathrm{Ext}_{\mathcal{P}}^{s,t}(\mathbb{F}_2, \mathbb{F}_2)$ has degree $(s, t, k, i) = (s, t, 0, 0)$ and q_n has degree $(s, t, k, i) = (0, 2^{n+1} - 1, 1, n)$. This is the algebraic Atiyah-Hirzebruch spectral sequence.

The alg-AHSS is relatively easy to analyze. For instance, it has a basis of the form $q_{i_1} \cdots q_{i_k} a$, where $a \in \mathrm{Ext}_{\mathcal{P}}(\mathbb{F}_2, \mathbb{F}_2)$. Moreover, the differential preserves the number of q_i, as the number of q_i is exactly the degree k.

We explain the principle to compute the alg-AHSS.

- For E_1 page, we know that $q_{i_1} \cdots q_{i_k} a$ becomes a linear independent basis. So we can write down generators for the E_1 page of alg-AHSS by counting out the basis.
- For differentials. It can be calculated within the cobar complex that computes the E_2 page of CESS. Therefore, we can use the $\mathcal{P}$-comodule structure to determine these differentials.

Example 2.6. We have $\psi(q_1) = \xi_1^2 \otimes q_0 + 1 \otimes q_1$. Since ξ_1^2 detects h_0 in $\mathrm{Ext}_{\mathcal{P}}^{1,2}(\mathbb{F}_2, \mathbb{F}_2)$, for any $a \in \mathrm{Ext}_{\mathcal{P}}(\mathbb{F}_2, \mathbb{F}_2)$ we have

$$d_1(q_1 a) = q_0 h_0 a$$

Example 2.7. We have $\psi(q_2) = \xi_2^2 \otimes q_0 + \xi_1^4 \otimes q_1 + 1 \otimes q_0$. The alg-AH differential d_r will reduce the i-degree by r, therefore taking the top degree we have:

$$d_1(q_2 a) = q_1 h_1 a$$

Suppose that $h_1 a = 0$. In this case we have $d_1(q_2 a) = 0$ and in the cobar complex $\xi_1^4 \otimes a$ becomes a coboundary. Therefore, on chain level we have

$$\begin{aligned}
\psi(q_2) \otimes a &= q_0 \otimes \xi_2^2 \otimes a + \xi_1^4 \otimes a \otimes q_1 \\
&= q_0 \otimes \xi_2^2 \otimes a + d(g) \otimes q_1 \\
&= q_0 \otimes \xi_2^2 \otimes a + g \otimes d(q_1) - d(g \otimes q_1) \\
&= q_0 \otimes (\xi_2^2 \otimes a + g \otimes \xi_1^2) - d(g \otimes q_1)
\end{aligned}$$

By taking the cohomology, we have

$$d_2(q_2 \otimes a) = q_0 \langle h_0, h_1, a \rangle$$

Example 2.8. More generally, we can determine higher differentials using the comodule structure and the cobar complex. In particular, for any $a \in \mathrm{Ext}_{\mathcal{P}}(\mathbb{F}_2, \mathbb{F}_2)$ we have:

(1) $d_1(q_{n+1}a) = q_n h_n a$.
(2) $d_2(q_{n+1}^2 a) = q_n^2 h_{n+1} a$.
(3) $d_2(q_{n+2}a) = q_n\langle h_n, h_{n+1}, a\rangle$ if $h_{n+1}a = 0$.
(4) $d_4(q_{n+2}^2 a) = q_n^2\langle h_{n+1}, h_{n+2}, a\rangle$ if $h_{n+2}a = 0$.
(5) $d_3(q_{n+3}a) = q_n\langle h_n, h_{n+1}, h_{n+2}, a\rangle$ if $h_{n+2}a = 0$ and $0 \in \langle h_{n+1}, h_{n+2}, a\rangle$.
(6) $d_6(q_{n+3}^2 a) = q_n^2\langle h_{n+1}, h_{n+2}, h_{n+3}, a\rangle$ if $h_{n+3}a = 0$ and $0 \in \langle h_{n+2}, h_{n+3}, a\rangle$.

These give us the principle to determine elements in the E_2 page of CESS.

Proposition 2.9. *Let $j \geq 5$. The E_2-page of the Cartan–Eilenberg spectral sequence takes the following form near h_j^3:*

$(s,\ k,\ t)$	$\mathrm{Ext}_{\mathcal{P}}^{s,t}(\mathbb{F}_2,\ \mathrm{Ext}_{\mathcal{Q}}^{k}(\mathbb{F}_2, \mathbb{F}_2))$
$(4,\ 1,\ 3\cdot 2^{j+1}+1)$	*contains* $\mathbb{F}_2\{q_0 g_{j-2}\}$
$(4,\ 2,\ 3\cdot 2^{j+1}+2)$	*contains* $\mathbb{F}_2\{q_0^2 g_{j-2}\}$
$(4,\ 3,\ 3\cdot 2^{j+1}+3)$	*contains* $\mathbb{F}_2\{q_0^3 g_{j-2}\}$
$(1,\ 2,\ 3\cdot 2^{j+1})$	0
$(1,\ 3,\ 3\cdot 2^{j+1}+1)$	0
$(1,\ 4,\ 3\cdot 2^{j+1}+2)$	0
$(1,\ 5,\ 3\cdot 2^{j+1}+3)$	0
$(3,\ 1,\ 3\cdot 2^{j+1}+1)$	$\mathbb{F}_2\{q_0 h_j^3\}$ *or* 0
$(3,\ 2,\ 3\cdot 2^{j+1}+2)$	$\mathbb{F}_2\{q_0^2 h_j^3\}$ *or* 0
$(3,\ 3,\ 3\cdot 2^{j+1}+3)$	$\mathbb{F}_2\{q_0^3 h_j^3\}$ *or* 0

Proof idea. By counting the degree of the basis we can determine the generators of the E_1 page in the above slots. The differentials given in Example 2.8 are enough to deduce the data in the proposition. □

Corollary 2.10. *Let $a = s + k$. There is no CE differentials entering $(a, t-a)$ degrees $(4, 3\cdot 2^{j+1}-4)$, $(5, 3\cdot 2^{j+1}-4)$, $(6, 3\cdot 2^{j+1}-4)$, $(7, 3\cdot 2^{j+1}-4)$.*

Proof. The CE-differential changes the tri-degree in the following way:

$$d_r \colon E_r^{a,s,t-a} = E_r^{s,k,t} \to E_r^{s+r,k-r+1,t} = E_r^{a+1,s+r,t-a-1}$$

The sparsity of CESS tells us that the odd $t-k$ degree vanish. So the only possible differentials entering the above tri-degrees are d_3 and d_5 by degree reason. All possible consequences are ruled out by the data in Proposition 2.9:

$d_r : E_r^{a,s,t-a} \to E_r^{a+1,s+r,t-a-1}$	argument
$d_3 : E_3^{3,1,3\cdot 2^{j+1}-3} \to E_3^{4,4,3\cdot 2^{j+1}-4}$	zero source
$d_3 : E_3^{4,1,3\cdot 2^{j+1}-3} \to E_3^{5,4,3\cdot 2^{j+1}-4}$	zero source
$d_3 : E_3^{5,1,3\cdot 2^{j+1}-3} \to E_3^{6,4,3\cdot 2^{j+1}-4}$	zero source
$d_3 : E_3^{5,3,3\cdot 2^{j+1}-3} \to E_3^{6,6,3\cdot 2^{j+1}-4}$	source all permanent cycles
$d_5 : E_5^{5,1,3\cdot 2^{j+1}-3} \to E_5^{6,6,3\cdot 2^{j+1}-4}$	zero source
$d_3 : E_3^{6,1,3\cdot 2^{j+1}-3} \to E_3^{7,4,3\cdot 2^{j+1}-4}$	zero source
$d_3 : E_3^{6,3,3\cdot 2^{j+1}-3} \to E_3^{7,6,3\cdot 2^{j+1}-4}$	source all permanent cycles
$d_5 : E_5^{6,1,3\cdot 2^{j+1}-3} \to E_5^{7,6,3\cdot 2^{j+1}-4}$	zero source

□

Now we can finish the proof of Theorem 2.1.

Proof of Theorem 2.1. We first prove the claim (1). The computation of [Che11] gives the desired result for data in degree $(a, t-a)$=$(4, 3\cdot 2^j-4)$, $(5, 3\cdot 2^j-4)$, $(3, 3\cdot 2^j-3)$, $(4, 3\cdot 2^j-3)$ and $(5, 3\cdot 2^j-3)$. It remains to show the following:

$(6, \quad 3\cdot 2^j-4)$	contains $\mathbb{F}_2\{h_0^2 g_{j-2}\}$
$(7, \quad 3\cdot 2^j-4)$	contains $\mathbb{F}_2\{h_0^3 g_{j-2}\}$

It suffices to show that the class $h_0^3 g_{j-2}$ is non-zero. This is because $h_0^3 g_{j-2}$ is detected by $q_0^3 g_{j-3}$ in CESS and the latter is not hit by any CE differential by Corollary 2.10.

We now prove (3). Using the isomorphism

$$\mathrm{Ext}_{\mathcal{A}^{\mathrm{mot}}}^{s+k,t,\frac{t-k}{2}}(\mathbb{F}_p[\tau], \mathbb{F}_p) \cong \mathrm{Ext}_{\mathcal{P}}^{s,t}(\mathbb{F}_2, \mathrm{Ext}_{\mathcal{Q}}^{k}(\mathbb{F}_2, \mathbb{F}_2))$$

we convert the mot-ASS into CESS. In the tridigree $(a, s, t-a) = (4, 4, 3\cdot 2^j-4)$, $(3, 3, 3\cdot 2^j-3)$ and $(5, 5, 3\cdot 2^j-3)$ we have $k = 0$. It follows from the computation of [Lin08] and [Che11] for $\mathrm{Ext}_{\mathcal{P}}^{s,2t} \cong \mathrm{Ext}_{\mathcal{A}}^{s,t}$. Therefore, by Proposition 2.9, it suffices to show the following data:

$(5, \quad 4, \quad 3\cdot 2^j-4)$	$\mathbb{F}_2\{h_0 g_{j-2}\}$
$(4, \quad 3, \quad 3\cdot 2^j-3)$	$\mathbb{F}_2\{h_0 h_j^3\}$
$(5, \quad 3, \quad 3\cdot 2^j-3)$	$\mathbb{F}_2\{h_0^2 h_j^3\}$

and for $j = 6$ there is possibly an additional class, $h_7 D_3(0)$, in degree $(a, s, t-a) = (5, 4, 3\cdot 2^6-4)$. These follows from a refined argument for the CESS E_2 page.

We begin with the first line. From Proposition 2.9, we know that there is a non-trivial element $q_0 g_{j-3}$ in degree $(a, s, t-a) = (5, 4, 3\cdot 2^j-4)$ detecting $h_0 g_{j-2}$. We claim that when $j \geq 7$ this is in fact the only non-trivial element. Suppose there exists another x besides $q_0 g_{j-3}$, this element is a permanent cycle by degree reasons and survive to the final page by Corollary 2.10. This will imply that the rank of degree $(a, t-a) = (5, 3\cdot 2^j-4)$ in the ASS E_2 page is at least 2, contradicting the calculation of [Lin08] and [Che11]. When $j = 6$, the rank of $(a, t-a) = (5, 3\cdot 2^j-4)$ is 2, implying there may be an additional element detecting $h_7 D_3(0)$.

For the second and the third line, it suffices to show that $h_0^2 h_j^3$ is not zero in CESS E_2 page. Suppose that this is not the case, then $h_0^2 h_j^3 \in \mathrm{Ext}_{\mathcal{A}}$ must be detected by other elements of

higher filtrations. This is impossible since the only choice is $(a, s, t-a) = (5, 5, 3\cdot 2^j - 3)$, where the only generator is $h_1 g_{j-3}$ that detects $h_1 g_{j-2}$(except for $j = 7$ we have an additional class $h_7 D(3)$ detecting $h_8 D(0)$).

The claim (2) and (4) follows from rigidity of Betti realization. □

2.3. **Determine alg-NSS differentials.** We now prove Theorem 2.2. Using Theorem 1.3, it is equivalent to show the following claim for alg-NSS differentials:

Theorem 2.11. *Using the name in CESS. There are following alg-NSS differentials:*

(1) $d_2(h_j^3) = 0$.
(2) $d_3(h_j^3) = 0$.
(3) $d_4(h_j^3) = q_0^3 g_{j-2}$.

Recall that the alg-NSS comes from the filtering by the augmented ideal I and takes the form

$$E_2^{s,k,t'} \cong \mathrm{Ext}^{s,t'}_{\mathrm{BP}_*\mathrm{BP}/I}(\mathrm{BP}_*/I, I^k/I^{k+1}) \Longrightarrow \mathrm{Ext}^{s,t'}_{\mathrm{BP}_*\mathrm{BP}}(\mathrm{BP}_*, \mathrm{BP}_*)$$
$$d_r \colon E_r^{s,k,t'} \to E_r^{s+1.k+r-1,t'}$$

Using the notation $\mathrm{BP}_*\mathrm{BP} \cong \mathrm{BP}_*[t_1, \cdots]$, we can compute the algebraic Novikov differentials in the cobar complex $\mathrm{cobar}(\mathrm{BP}_*\mathrm{BP})$ through the following procedure:

- Suppose we have a class ϵ in the E_2 page. We want to compute the differential $d_*(\epsilon)$. For simplicity, we assume that the k degree of ϵ is zero. Otherwise, we need only to multiply each cobar complex with I^j, where j is the k degree of ϵ.
- We pick a cocycle e in $\mathrm{cobar}(\mathrm{BP}_*\mathrm{BP}/I)$ that represents ϵ.
- We choose a lift e_2 of e in $\mathrm{cobar}(\mathrm{BP}_*\mathrm{BP})$. Note that e_2 may not be a cocycle.
- We calculate $c_2 = d(e_2)$. Since e_2 is a cocycle modulo I, we have $c_2 \in I \cdot \mathrm{cobar}(\mathrm{BP}_*\mathrm{BP})$.
- We begin to iterate. Now we have $e_r \in \mathrm{cobar}(\mathrm{BP}_*\mathrm{BP})$ and $c_r = d_r(e_r)$. We also have $c_r \in I^{r-1} \cdot \mathrm{cobar}(\mathrm{BP}_*\mathrm{BP})$. We break into two cases:
  (1) Suppose that we can find $w_r \in I^{r-1} \cdot \mathrm{cobar}(\mathrm{BP}_*\mathrm{BP})$ such that $d(w_r) = c_r \bmod I^r$, this means c_r becomes a coboundary in $I^{r-1} \cdot \mathrm{cobar}(\mathrm{BP}_*\mathrm{BP})/I^r$. Therefore, we conclude that $d_r(\epsilon) = 0$ and continue with $e_{r+1} = e_r + w_r$.
  (2) If such w_r does not exist, then we conclude that there is a class
  $$\zeta \in \mathrm{Ext}^{*,*}_{\mathrm{BP}_*\mathrm{BP}/I}(\mathrm{BP}_*/I, I^{r-1}/I^r)$$
  which is detected by c_r and we have $d_r(\epsilon) = \zeta$.

This gives a general procedure to produce differentials. We note that, the crucial part in the above procedure is to find suitable w_r and to find the correct class that is detected by c_r. Therefore, in order to make use of the above procedure to prove Theorem 2.11, we need some preparation:

Definition 2.12. Let T_j denote the following class in $\mathrm{cobar}(\mathrm{BP}_*\mathrm{BP})$:

$$T_j := \frac{1}{2} d(t_1^{2^j})$$

Note that T_j is a cocycle. We let ϑ_j denote the class in the ANSS E_2 page that is detected by T_j. In fact, under the Thom reduction, ϑ_j is sent to h_j^2 in the Adams E_2 page.

Lemma 2.13. *Let $j \geq 5$, the product $\vartheta_j \vartheta_{j+1}$ in the ANSS E_2 page is detected by $q_0^2 g_{j-2}$ in alg-NSS.*

Proof of Theorem 2.11. We follow the procedure to determine the alg-NSS differentials on h_j^3. We first need to choose a lift of h_j^3 to $\mathrm{cobar}(\mathrm{BP}_*\mathrm{BP})$. This class is given by

$$e_2 = t_1^{2^j}|T_{j+1}$$

Indeed, under the identification $\mathrm{BP}_*\mathrm{BP}/I \cong \mathcal{A}$ the element $t_1^{2^j}$ detects h_j and T_{j+1} detects ϑ_{j+1} which is sent to h_j^2. Calculating the cobar differentials, we have

$$c_2 = d(e_2) = -2T_j|T_{j+1} \in I \cdot \mathrm{cobar}(\mathrm{BP}_*\mathrm{BP})$$

In order to proceed on, we use Lemma 2.13. The product $\vartheta_j\vartheta_{j+1}$ is detected by $q_0^2 g_{j-2}$. This means, we have a correction term c such that

$$T_j|T_{j+1} + d(c) \in I^2 \cdot \mathrm{cobar}(\mathrm{BP}_*\mathrm{BP})$$

whose image in the cohomology of I^2/I^3 detects $q_0^2 g_{j-2}$. Therefore, we can choose $w_2 = -2c \in I \cdot \mathrm{cobar}(\mathrm{BP}_*\mathrm{BP})$. We have

$$c_2 - d(w_2) = -2T_j|T_{j+1} - 2d(c) \in I^3 \cdot \mathrm{cobar}(\mathrm{BP}_*\mathrm{BP})$$

This proves $d_2(h_j^3) = 0$.

We continue. The procedure tells us that we have

$$e_3 = e_2 + w_2 = t_1^{2^j}|T_{j+1} - 2c$$
$$c_3 = d(e_3) = -2T_j|T_{j+1} - 2d(c)$$

Since c_3 is already in I^3, we simply choose $w_3 = 0$. This proves $d_3(h_j^3) = 0$. Next, we have

$$e_4 = e_3 = e_2 + w_2 = t_1^{2^j}|T_{j+1} - 2c$$
$$c_4 = c_3 = d(e_3) = -2T_j|T_{j+1} - 2d(c)$$

Since c_4 detects $q_0^3 g_{j-2}$, we conclude that $d_4(h_j^3) = q_0^3 g_{j-2}$. □

Remark 2.14. In the above proof, we do not discuss the existence of w_4. Therefore, the above discussion can not conclude that $q_0^2 g_{j-2}$ is nonzero.

We quickly explain the proof of Lemma 2.13. The formal proof is too long for this talk. It is very complicated and with a little coding involved in. The spirit is, we can choose a special c_j such that $T_j|T_{j+1} + d(c_j)$ is zero after modulo $(4, v_1^4)$. Therefore, this cocycle detects some class $q_0^2 H$ for $H \in \mathrm{Ext}_{\mathcal{A}}$. We can identify H with g_{j-2} using an inspection of May filtration terms.

2.4. **Determine classical ASS differentials.** We now prove Theorem 2.3. In order to determine these differentials, we will make use of $\mathbb{F}_2$-synthetic homotopy theory.

We use $\mathbb{S}^{k,s}$ to denote the bigraded sphere $\Sigma^{-s}\nu S^{k+s}$ in the category $\mathrm{Syn}_{\mathbb{F}_2}$. $\pi_{k,s}(X)$ is defined to be the Abelian group of homotopy classes from $\mathbb{S}^{k,s}$ to X. The canonical comparison map gives

$$\lambda \colon \mathbb{S}^{0,1} = \Sigma^{-1}\nu S^1 \to \nu\Sigma^{-1}S^1 = \mathbb{S}^{0,0}$$

specifies a homotopy class.

Lemma 2.15 ([Pst23]; [BHS23])**.** *Let $X \in \mathrm{Sp}$. There is a natural isomorphism of bigraded Abelian groups*

$$\pi_{t-s,s}(\mathbb{S}/\lambda \otimes \nu X) \cong \mathrm{Ext}_{\mathcal{A}}^{s,t}(\mathbb{F}_2, (\mathbb{F}_2)_* X)$$

Theorem 2.16 (Theorem 8.19; [BHS23])**.** *Let $X \in \mathrm{Sp}$ be $\mathbb{F}_2$-complete. Let x denote a class in the ASS E_2 page of X.*

- *In the point of view of Lemma 2.15, the followings are equivalent:*
  (1) $d_2, \cdots, d_r$ vanish on x.
  (2) $x \in \pi_{t-s,s}(\mathbb{S}/\lambda \otimes \nu X)$ can be lifted to $\tilde{x} \in \pi_{t-s,s}(\mathbb{S}/\lambda^r \otimes \nu X)$.
  In this case, we can choose a lift $\tilde{x} \in \pi_{t-s,s}(\mathbb{S}/\lambda^r \otimes \nu X)$ such that $d_{r+1}(x)$ is identified with $r(\tilde{x})$, where r is the λ-Bockstein:
$$\mathbb{S}/\lambda^r \otimes \nu X \to \mathbb{S}/\lambda \otimes \nu X$$
- *Suppose additionally that x is a permanent cycle. Then for any lifting $\tilde{x}$ we have:*
  (a) If x survives to E_{r+1} page, then $\lambda^{r-1}\tilde{x} \neq 0$.
  (b) If x survives to E_∞ page, then the image of $\tilde{x}$ in $\pi_k(X)$ is detected by x.
- *Moreover, for a permanent cycle x, we can always find a lifting $\tilde{x}$ that satisfies the followings:*
  (1) If x is a target of a d_{r+1} differential, then we can choose $\tilde{x}$ such that $\lambda^r\tilde{x} = 0$.
  (2) If x survives to the E_∞ page, and $\alpha \in \pi_k(X)$ is detected by x. Then we may choose $\tilde{x}$ such that $\lambda^{-1}\tilde{x} = \alpha$.

This brings a way for us to discuss vanishings of differentials. In particular, now we can prove Theorem 2.3.

Lemma 2.17. *Consider the Bockstein sequence:*
$$\Sigma^{0,-1}\mathbb{S}^{0,0}/\lambda^{r-1} \xrightarrow{\kappa_r} \mathbb{S}^{0,0}/\lambda^r \xrightarrow{f_r} \mathbb{S}^{0,0}/\lambda \xrightarrow{\delta_r} \Sigma^{1,-1}\mathbb{S}^{0,0}/\lambda^{r-1}$$
Then $\delta_4(h_j^3)$ is divisible by $\tilde{h}_0^2$, where $\tilde{h}_0$ is a synthetic homotopy class that is detected by h_0.

This lemma is concluded from three things:

- We have a lift of h_j^2 to $\theta_j \in \pi_{*,*}(\mathbb{S}^{0,0}/\lambda^4)$. Consequently we have $\delta_4(h_j^3) = \theta_j\delta_4(h_j)$. This part follows from a synthetic refinement of the inductive approach of [BJM83].
- We have $\delta_4(h_j) = \tilde{h}_0\theta_{j-1}$. This follows from an inspection of Bockstein over $\pi_{*,*}(\mathbb{S}^{0,0}/\lambda)$.
- We have $\theta_j\theta_{j-1}$ divisible by $\tilde{h}_0^2$. This follows from a reshuffle of Toda bracket:
$$\theta_j\theta_{j-1} = \langle \tilde{h}_0, \lambda\theta_{j-1}, \theta_{j-1}, 2\rangle\theta_j = \tilde{h}_0\langle \lambda\theta_{j-1}, \theta_{j-1}, 2, \theta_j\rangle$$

Proof of Theorem 2.3. Using Theorem 2.16, the vanishing of higher differentials on h_j^3 is equivalent to the existence of liftings in $\pi_{*,*}(\mathbb{S}/\lambda^r)$. While the obstruction of the lifting is the image $\delta_r(h_j^3)$. Consider the cofiber sequence:
$$\mathbb{S}^{0,0}/\lambda^4 \xrightarrow{r_4} \mathbb{S}^{0,0}/\lambda \xrightarrow{\delta_4} \Sigma^{1,-1}\mathbb{S}^{0,0}/\lambda^3$$
By Lemma 2.17, we must have $\delta_4(h_j^3) = \tilde{h}_0^2 y$ for some $y \in \pi_{3\cdot 2^j-4,3}\mathbb{S}^{0,0}/\lambda^3$. One can show by Lemma 2.15 that $\pi_{3\cdot 2^j-4,3}\mathbb{S}^{0,0}/\lambda^3 = \mathbb{Z}/4\{\kappa[g_{j-2}]\}$, where $[g_{j-2}]$ is a lifting of g_{j-2} to $\mathbb{S}/\lambda^2$. Therefore, there are three cases:

- If $y = 0$, this means the obstruction vanishes so h_j^3 lifts to $\mathbb{S}^{0,0}/\lambda^4$, which is equivalent to say that $d_2(h_j^3) = 0, d_3(h_j^3) = 0, d_4(h_j^3) = 0$.
- If $y = 2\kappa[g_{j-2}]$, then after apply r, y becomes zero. Therefore, we have $d_2(h_j^3) = 0, d_3(h_j^3) = 0, d_4(h_j^3) = h_0^3 g_{j-2}$.
- If y is a generator, then we only have $d_2(h_j^3) = 0$ and $d_3(h_j^3) = h_0^2 g_{j-2}$.

□

References

[BHS23] Robert Burklund, Jeremy Hahn, and Andrew Senger. "On the boundaries of highly connected, almost closed manifolds". In: *Acta Mathematica* 231.2 (2023), pp. 205–344 (cit. on p. 12).

[BJM83] MG Barratt, JDS Jones, and ME Mahowald. "The Kervaire invariant problem". In: *Proceedings of the Northwestern Homotopy Theory Conference (Evanston, Ill., 1982)(Providence, RI), Contemp. Math.* Vol. 19. 1983, pp. 9–22 (cit. on p. 13).

[BX25] Robert Burklund and Zhouli Xu. "The Adams differentials on the classes h_j^3". In: *Inventiones mathematicae* 239.1 (2025), pp. 1–77 (cit. on p. 1).

[Che11] Tai-Wei Chen. "Determination of $\mathrm{Ext}_{\mathcal{A}}^{5,*}(\mathbb{Z}/2,\mathbb{Z}/2)$". In: *Topology and its Applications* 158.5 (2011), pp. 660–689 (cit. on p. 10).

[Ghe18] Bogdan Gheorghe. "The Motivic Cofiber of *tau*". In: *Documenta Mathematica* 23 (2018), pp. 1077–1127 (cit. on p. 4).

[GWX21] Bogdan Gheorghe, Guozhen Wang, and Zhouli Xu. "The special fiber of the motivic deformation of the stable homotopy category is algebraic". In: *Acta Mathematica* 226.2 (2021), pp. 319–407 (cit. on p. 4).

[HKO11] Po Hu, Igor Kriz, and Kyle Ormsby. "Remarks on motivic homotopy theory over algebraically closed fields". In: *Journal of K-Theory* 7.1 (2011), pp. 55–89 (cit. on p. 3).

[HKØ17] Marc Hoyois, Shane Kelly, and Paul Østvær. "The motivic Steenrod algebra in positive characteristic." In: *Journal of the European Mathematical Society (EMS Publishing)* 19.12 (2017) (cit. on p. 4).

[Lin08] Wen-Hsiung Lin. "$\mathrm{Ext}_{\mathcal{A}}^{4,*}(\mathbb{Z}/2,\mathbb{Z}/2)$ and $\mathrm{Ext}_{\mathcal{A}}^{5,*}(\mathbb{Z}/2,\mathbb{Z}/2)$". In: *Topology and its Applications* 155.5 (2008), pp. 459–496 (cit. on p. 10).

[MVW06] Carlo Mazza, Vladimir Voevodsky, and Charles A Weibel. *Lecture notes on motivic cohomology*. Vol. 2. American Mathematical Soc., 2006 (cit. on p. 3).

[Pst23] Piotr Pstragowski. "Synthetic spectra and the cellular motivic category". In: *Inventiones mathematicae* 232.2 (2023), pp. 553–681 (cit. on p. 12).

[Rav03] Douglas C Ravenel. *Complex cobordism and stable homotopy groups of spheres*. American Mathematical Soc., 2003 (cit. on p. 2).

[Vez01] Gabriele Vezzosi. "Brown-Peterson spectra in stable $mathbbA_1$-homotopy theory". In: *Rendiconti del Seminario Matematico della Universita di Padova* 106 (2001), pp. 47–64 (cit. on p. 3).

[Voe03] Vladimir Voevodsky. "Reduced power operations in motivic cohomology". In: *Publications Mathématiques de l'IHÉS* 98 (2003), pp. 1–57 (cit. on p. 4).